GOAL

The goal of each player is to reach a certain amount of points of carbon, nitrogen, and phosphorus as fast as possible (adapted from the original card game) while managing these same points for survival.

CARD ARCHETYPES

1) **ORGANISMS** — 16 Cards - Specifically plants, algae, detritivores (shredders, collectors, grazers, etc.), and consumers/predators. These cards are to consume or to be consumed but also the main source of nutrients. Can die to insufficient nutrients, and also has evolution/adaptation mechanics.

    a. **PRODUCERS**: Plant, Algae, Etc. — 7 Cards
    Consumes nutrients— mainly carbon and nitrogen to produce oxygen (O_2). Has a fixed amount of lifespan / rounds. Can be eaten by detritivore cards played by any player.

    b. **DETRITIVORES**: Collectors, Grazers, Shredders — 6 Cards
    Can eat alive and dead plant matter (regardless from which player), releasing small amounts of nutrients per round. Can not fully consume a producer card in one round, instead it may take multiple rounds depending how much detritivores are eating it.

    c. **CONSUMER**: Secondary — 3 Cards
    Specifically secondary consumers— when consuming detritivores/primary consumers, the biomass of the organism transfers to the consumer. The nutrient pool of the consumer will be larger, allowing more nutrients available to be released when decomposing. Additionally, it can be used to protect and attack against enemy cards.

2) **MICROBE —** 6 Cards - Bacteria/Fungi that can release all or a specific nutrient during decomposition. Allows faster decomposition done by detritivores, returning more nutrients to be back for use. Can only be used on 1 dead card and can not be transferred or retracted back into the player's hand. Does not consume collected nutrient points (except for oxygen), but reduces net nutrient gain directly from the dead card.

3) **WILD EVENT** — 4 Cards splitted to 2 Decks - Random events that will occur to both players when drawn. Results can be advantageous or not to both players. When drawn, the event will immediately be played but will not consume any action points.

4) **EVOLUTION / ADAPTATION —** 10 Cards - Enhances the abilities of organisms. Must complete required conditions for activation.

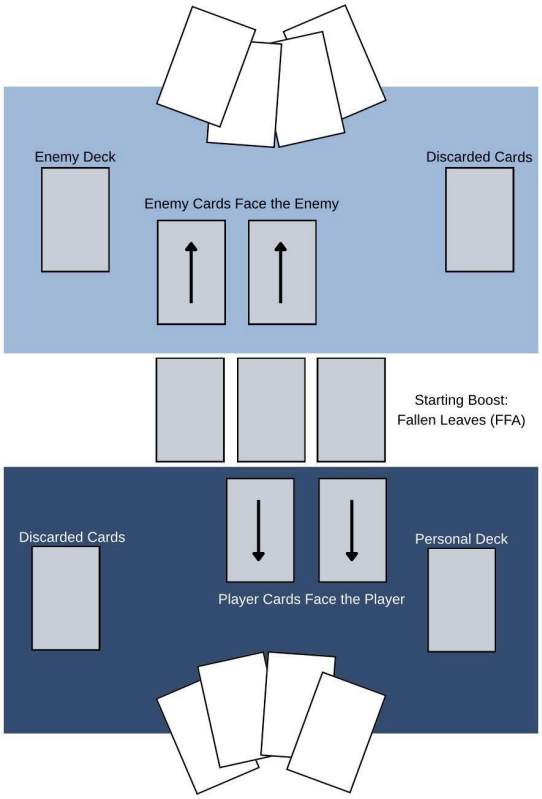
RULES & MECHANICS

(A) Player Deck, Hand, Rounds & Turns

- A deck containing 16 Organism Cards, 6 Microbe Cards, 10 Evolution Cards, and 4 (splitted into 2 decks) Wild Event Cards are shuffled into separated decks for each players — totaling 36 Cards overall for each player
- Each player will start with a hand of 5 random cards
- A player has 3 action points (fixed) per round
- A card can be disposed of for a new card, costing 1 point. Cards may also be held for future rounds
- Upon placement of any cards (except wild event cards), they consume 1 action point.
- When cards are placed/played, it CAN NOT be retracted (except for bacteria cards)
- Disposed cards go to a separate disposed deck, which can only be reused or reshuffled when the first deck is fully depleted.

(B) Game Set-up

- No more fallen leaves



(C) Card-Specific Mechanics

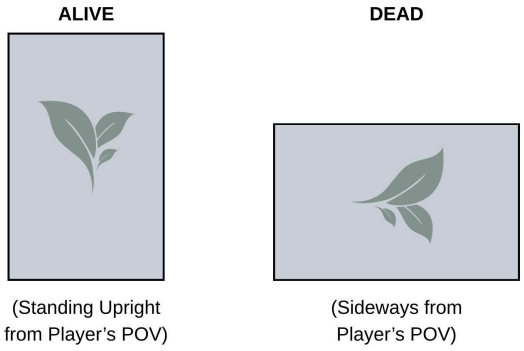
(C.1.) Organism Cards have 2 statuses: Alive & Dead

Alive - If activation conditions are satisfied, organisms will do specific processes. They may either die to the lack of necessary nutrients points

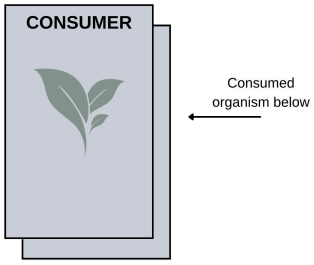
or to natural causes via a number of rounds or wild event cards. Can be consumed by other selected organisms.

Dead / **Decaying Matter** - Organisms that can be readily be consumed by detritivore cards. Using a detritivore card will separately cost another action point in the current round but will not in the succeeding rounds. Is not immediately depleted in a single round. Complete decomposition will take multiple rounds depending on the detritivores and microbes.

(C.2.) Placement indication for the organism cards:

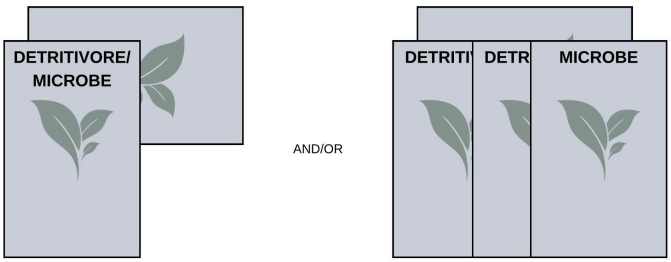


(C.3.) Consumer-Specific Interactions— indications:



- A consumed organism stays below the consumer card but stays visible for tracking purposes.

(C.4.) Detritivore and Microbe-Specific Interactions— indication:



- The decaying matter getting broken down stays below any microbe or detritivore card. It must stay visible even with stacked cards for tracking purposes.

(D) Point Management System

- There are 4 kinds of nutrient points to be used and earned to the player's advantage. Three (3)— carbon, nitrogen, and phosphorus— are necessarily needed to win. One (1)— oxygen is essential for managing your cards.
- The player is given starting nutrients:
  - 10 Carbon ©
  - 3 Nitrogen (N)
  - 0 Phosphorus (P)
  - 7 Oxygen (O)
- Oxygen is not needed to win however it is used for the majority of the abilities/activation conditions.
- Points can go beyond the winning condition, but may result to consequences.

(E) Winning Conditions

- A player must achieve:
  - 50 Carbon © Points
  - 15 Nitrogen (N) Points
  - 1 Phosphorus (P) Points
- A digital sheet for tracking these points will be given beforehand

PRODUCER CARDS

Oxygen Production is per day round due to active production/photosynthesis. The ratio of carbon to oxygen at day is 1:1 except for Algae Cards

Cost refers to activation conditions, nutrients are only consumed during the first play.

Production is the net gain after a round, may contribute to player's nutrient score and/or the organisms' biomass.

Biomass is the nutrients that will be available for decomposition and nutrient cycling after dying

Plant Card	C Cost	N Cost	Lifespan	Biomass	Production to Biomass (per Round)	Production to Nutrient Pool (per Round)
Emergent Plant	-2	-0.5	3 rounds	6 C 1 N 0 P	1 C	2 O 1 C
River Grass	-2	-0.5	4 rounds	6 C 1 N 0 P	1 C	2 O 1 C
Aquatic Plant	-2	-0.5	4 rounds	6 C 1 N 0 P	1 C	2 O 1 C
Free-floating Plant	-2	-0.5	3 rounds	6 C 1 N 0 P	1 C	2 O 1 C

Algal blooms will happen if phosphorus points go past 2

Algae Card	C Cost	N Cost	Lifespan	Biomass	Production to Biomass (per Round)	Production to Nutrient Pool (per Round)
Green Algae	-1	-0.5	2 rounds	2 C 0 N 0 P	1 C 1 N	3 O 1 C

Diatoms	-2	-0.5	1 rounds	2 C 0 N 0 P	1 C 1 N	4 O 1 C
Cyanobacteria	-1	-0.5	2 rounds	2 C 0 N 0 P	1 C 1 N	3 O 1 C 1 N

Example Scenario.
Emergent plant successfully photosynthesizes for a round

-> +1 C to Biomass -> Biomass becomes 7 C, 1 N, 0 P next round

-> +1 C to Nutrient Score

DETRITIVORE CARDS

Shredders —> Break large dead materials
Collectors —> Consume fine organic matters
Grazers —> Consume algae/biofilm

Lifespan of detritivores can be extended for the total of rounds they are consuming dead matter. Biomass will not be added nor deducted.

Oxygen Consumption is per round. Nitrogen is a one-time use upon activation.

Shredder Cards	O Cost	N Cost	Lifespan	Biomass	Ability
Freshwater Shrimp	-1	-0.5	4 rounds	3 C 1 N 0 P	**Shred:** Exclusively eats dead plant cards. Releases 3 C from a dead plant's biomass per round
Aquatic Worm	-1	-0.5	3 rounds	3 C 1 N 0 P	**Burrow:** Exclusively eats dead plant cards. Releases 2 C from a dead plant's biomass per round

Collector Cards	O Cost	N Cost	Lifespan	Biomass	Ability
Tubifex Worm	-1	-0.5	3 rounds	3 C 1 N 0 P	**Filter:** When consuming a dead plant or dead algae card, release 1 C from their biomass. When paired with a Shredder, releases an additional carbon along with a nitrogen point
Chrinomid Larva	-1	-0.5	3 rounds	3 C 1 N 0 P	**Fine Detritus:** When processing a dead plant or dead algae card, release 1 C from their biomass. When paired with a Shredder, releases an additional carbon along with a nitrogen point

Grazer Cards	O Cost	N Cost	Lifespan	Biomass	Ability

Freshwater Snail	-1	-0.5	4	3 C 1 N 0 P	**Graze:** Exclusively eats algae cards (both alive and dead). Releases 1 carbon from their biomass. Can prevent algal blooms
Aquatic Grazer	-1	-0.5	3	3 C 1 N 0 P	**Control Algae:** When consuming algae, remove 1 excess P associated with that algae. Manages phosphorus level before algal bloom becomes a problem.

MICROBE CARDS

- Return nutrients from dead organic matter to the nutrient pool.
- After assisting with decomposing, returns back to the player's hand

Card	O Cost	Main ability
Carbon Bacteria ×2	-1	When recycling nutrients along with a detritivore, released carbon by detritivore is doubled but does not affect the biomass of the dead card.
Nitrifying Bacteria x2	-1	When recycling nutrients along with a detritivore, it releases all nitrogen from the dead card.
Phosphorus Bacteria x2	-1	When decomposing Processed Detritus, gain 1 P.

Wild Event Cards

Wild Event	Trigger / Condition	Effect	Ecological Concept
Strong Rainfall	Played during Environmental Phase	All players gain **+1 P**. If a player has more than 2 **P** and at least 1 Algae Card, trigger **Algal Bloom**. Algae Cards Biomass is doubled	Nutrient runoff → eutrophication → algal bloom → oxygen depletion
Drought	Played during Environmental Phase	During the round, Producers produce **−1 O_2**. Aquatic organisms cannot be played. Aquatic Plants cannot photosynthesize.	Water limitation → reduced productivity and habitat availability
Heat Wave	Played during Environmental Phase	All organisms consume **+1 O_2**. Organisms that cannot meet their O_2 requirement die.	Higher temperature → increased oxygen stress / lower oxygen availability
Flood	Played during Environmental Phase	A random organism card that is already played will die. Player gains **+1 C**. If no organism is available, gain **+1 P** instead.	Flooding → physical disturbance + transport of organic matter and nutrients